

Instructions to the Students:

1. Solve ANY FIVE questions out of the following.
2. Draw figures wherever necessary.
3. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve the following.

CO01

- A) In the circuit shown in Fig. 1, determine current through 5 ohm resistance using mesh analysis.

[06]

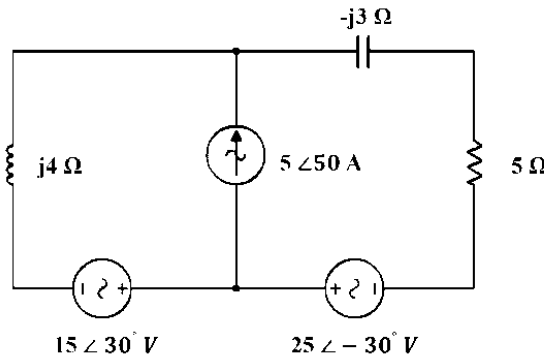


Fig. 1

- B) In the circuit shown in Fig. 2, obtain the value of the load impedance between the terminals X and Y for maximum power transfer. Hence, determine the maximum power absorbed by this load.

[06]

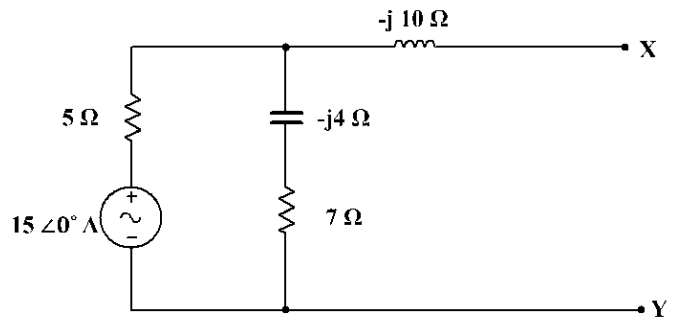


Fig. 2

Q.2 Solve Any Two of the following.

**CO01,
CO02**

- A) Derive expression for frequencies at which voltage across L and C are maximum in a series resonant circuit. Also determine values of voltages across L and C at this frequency. [06]
- B) A coil of 12H and resistance of 15Ω is shunt with 100pf capacitor. The combination is connected across a generator of 100V, having internal resistance of $70k\Omega$, Determine [06]
(a) voltage across parallel circuit at resonance and
(b) bandwidth.
- C) Derive an expression for selectivity and bandwidth of anti-resonant circuit. If the circuit impedance of a parallel resonating frequency is $(100 + j10)\Omega$ at frequency 2.5 MHz. Find value of L and C if the Q of inductor is 6 and it is constant. [06]

Q.3 Solve the following.

**CO01,
CO02**

- A) For a symmetrical T network, explain briefly the terms, [08]
a) characteristic impedance
b) Propagation constant.
Also derive and expressions for these parameters in terms of circuit impedances.
- B) Design a prototype low pass filter sections so as to have design impedance of 700Ω and $f_c = 2\text{kHz}$. Find circuit elements. Also find characteristic impedance Z_o at frequency of 1kHz, 2kHz and 5kHz. [04]

Q.4 Solve any two of the following.

CO01

- A) For a 2-port passive network, define- [06]
(a) driving point impedance
(b) driving point admittance
(c) current transfer ratio
(d) transfer admittance
- B) For the network shown in Fig. 3 below, determine Z (open circuit impedance) parameter. Also verify condition for reciprocity and symmetry for the same. [06]

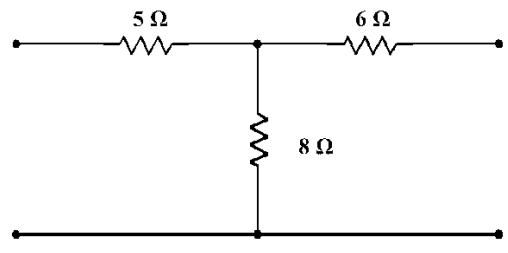


Fig. 3

- C) Obtain equations for Y-parameters in terms of Z, h and ABCD parameters. [06]

Q. 5 Solve the following.

CO01,CO03

- A) In the R-C circuit shown in below Fig 4, the switch is closed at $t = 0$. Obtain expression for current $i(t)$. [06]

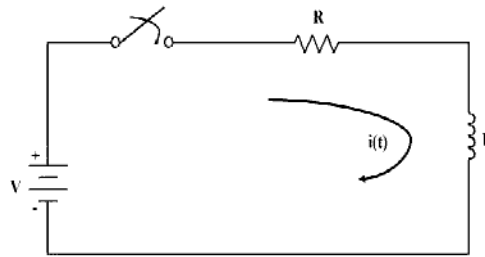


Fig 4

- B) In the circuit shown in Fig.5, the switch K is moved from position 'a' to position 'b' at $t = 0$. A steady state having previously been established at when the switch was position 'a'. Using Laplace Transform method, find value of $i(t)$. [06]

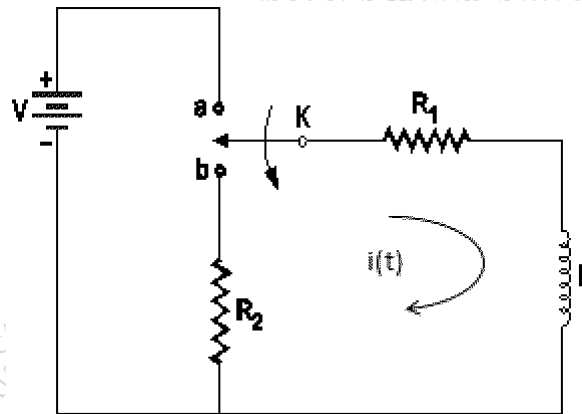


Fig. 5

Q. 6 Solve the following.

**CO03,
CO04**

- A) For a transmission line of length l , starting from the differential equations, derive the expressions for the sending-end voltage and current in terms of receiving end voltage and currents and the secondary line constants. [06]
- B) A $50 \, \Omega$ lossless transmission line of length $1.37 \, \lambda$ is terminated into load of $(60 + j 40) \, \Omega$. Using Smith Chart, Find [06]
- input impedance of the line
 - input and load admittances
 - Reflection coefficient in amplitude and phase
 - Standing wave ratio (S)
 - Minimum and maximum values of the impedances along the line
 - Distance of 1st voltage minima and maxima from the load

***** End *****